MIL-P-63137B (AR)
18 DECEMBER 1981
SUPERSEDING
MIL-P-63137A (AR)
1 February 1979

MILITARY SPECIFICATION

PROJECTILE, 8-INCH, HE, RA, M650
WARHEAD SECTION, FORWARD

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Warhead Section, Forward, for use in Projectile, 8-Inch, HE, RA, M650.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this Military Specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	-Cleaning Methods and Pre-treatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-H-6088	-Heat Treatment of Aluminum Alloys

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-141 -Paint, Varnish, Lacquer and Related Materials; Method of Inspection and Testing

MILITARY

MIL-STD-1169 -Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

ARRADCOM

9287954 -Warhead Section, Forward

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E8 -Tension Testing of Metallic Materials

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

ANSI B46.1 -Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, New York 10078).

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings, specifications and standards.

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3.2 Warhead section, forward. The warhead section, forward, shall comply with all requirements specified on Drawing 9287954 and with all requirements specified in applicable specifications and standards.

3.3 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

3.4 Metal defects. The warhead section, forward, shall be free from cracks, laps, seams, ruptures, pits and other open surface discontinuities.

3.5 Visual standards. Visual standards for metal defects will be established for each contract after the first article or whenever requested by the procuring contracting agency.

3.6 Heat treatment. Heat treatment equipment, procedures and controls shall be in accordance with MIL-H-6088. The forgings shall be solution heat treated in such a manner as to avoid overheating of the material or cracking during quenching.

3.7 First article inspection. This specification makes provisions for a first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.8.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.8.2 Finishing. Finishing shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment.

3.8.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of specification MIL-A-2550.

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3.8.4 Burr. No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.8.5 Foreign matter. No part shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Warhead Section Forward	9287954	15

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63137B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Warhead Section, Forward				9287954
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Warhead Section, Forward (Dwg. 9287954)				
	Examination for Defects	15	-	3.2	4.4.2.1
	Salt Spray	2	-	3.2	4.4.3.1.1
	Tension Test	1	-	3.2	4.4.3.2

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition to these provisions, the lot formation provisions of 4.4.1.1 shall apply.

4.4.1.1 Heat treat lot. A heat treat lot shall contain a maximum of 1200 forgings, heat treated continuously and/or consecutively in an unchanged process.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63137B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9287954
4.4.2.1	Warhead Section, Forward				NEXT HIGHER ASSEMBLY 9288021
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal Defects		100%	3.4	Visual
MAJOR					
101	Pitch diameter of internal thread		0.40%	3.2	Gage
102	Minor diameter of internal thread		0.40%	3.2	Gage
103	Pitch diameter of external thread		0.40%	3.2	Gage
104	Major diameter of external thread		100%	3.2	Gage
105	Location of small basic diameter		0.40%	3.2	Gage
106	Runout of small basic diameter		0.40%	3.2	Gage
107	Location of large basic diameter		0.40%	3.2	Gage
108	Runout of large basic diameter		0.40%	3.2	Gage
109	Length of internal thread including undercut		0.40%	3.2	Gage
110	Runout of pitch diameter of internal thread, including seating of gage on shoulder (2 locations)		0.40%	3.2	Gage
111	Wall thickness of taper		0.40%	3.2	Gage
112	Perpendicularity of shoulder with pitch diameter of external thread		0.40%	3.2	Gage
113	Large inside diameter		0.40%	3.2	Gage
114	Next largest inside diameter		0.40%	3.2	Gage
115	Second smallest inside diameter in nose		0.40%	3.2	Gage
116	True position of second smallest inside dia.		0.40%	3.2	Gage
117	Distance from nose to end of second smallest I.D.		0.40%	3.2	Gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Warhead Section, Forward				9287954
					NEXT HIGHER ASSEMBLY
					9288021
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
118	Distance from nose to end of third smallest I.D.		0.40%	3.2	Gage
119	Length of external thread including undercut		0.40%	3.2	Gage
120	Tension Test	A.		3.2	4.5.2
MINOR					
201	Length from shoulder to nose end		0.65%	3.2	Gage
202	Length of external thread undercut		0.65%	3.2	Gage
203	Length of internal thread undercut		0.65%	3.2	Gage
204	Protective finish improper or missing		0.65%	3.2	4.5.1
205	Marking missing or unidentifiable		0.65%	3.2	Visual
206	Chamfer missing on exterior and interior threads		0.65%	3.2	Visual
207	Surface finish improper		0.65%	3.2	Visual
208	Diameter of nose thread undercut		0.65%	3.2	Gage
209	Length of largest inside diameter		0.65%	3.2	Gage
210	Small inside diameter in nose		0.65%	3.2	Gage
211	True position of small inside diameter in nose		0.65%	3.2	Gage
212	Distance from nose to end of small inside diameter		0.65%	3.2	Gage
213	Third smallest inside diameter in nose		0.65%	3.2	Gage
214	True position of third smallest diameter in nose		0.65%	3.2	Gage
215	Painted surface with total exposed or damaged area in excess of 1/4 sq. in.		0.65%	3.2	Visual
216	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

A. One (1) sample to be selected randomly from each heat treat lot.

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4.4.3 Testing.

4.4.3.1 Cleaning and application of protective coating. The sample and size and acceptable criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the action recommended therein.

4.4.3.1.1 Salt spray. Salt spray samples shall be selected alternately, one during the first 30 minutes of the day's production and one randomly at different times of the day. Failure to comply with the requirements of the applicable drawing shall be cause for taking the action recommended by the applicable specification.

4.4.3.2 Tension Test. One forging shall be randomly selected from each heat treat lot and subjected to the tension test in accordance with 4.5.2. The forward warheads selected for tensile testing shall have two specimens taken, one each, from the locations shown on Dwg. 9287954. Failure to meet the required properties shall be cause for rejection of the heat treat lot represented. The rejected heat treat lot may, at the option of the contractor, be reheat treated to form a new heat treat lot.

4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Protective coating. The procedures for testing the cleaning and the surface treatment methods for components shall be in accordance with the method and criterial for testing as specified in the applicable specification.

4.5.1.1 Salt spray. Salt spray testing shall be accomplished on production warheads or sectioned warheads that were acceptable prior to painting. If sectioned warheads are used, sectioning shall be accomplished to allow testing of all major areas on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by Federal Test Standard No. 141 and TT-C-490.

4.5.2 Tension testing. Testing shall be in accordance with ASTM Specification E8.

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5. PREPARATION FOR DELIVERY.

5.1 Packaging. The warhead section, forward, shall utilize packaging in accordance with the contract and MIL-STD-1169, Level B.

5.2 Packing and marking. The warhead section, forward, shall be packed and marked in accordance with the contract and MIL-STD-1169, Level B.

6. NOTES.

6.1 Intended use. The components covered by this specification are intended for use on the 8-Inch, M650, RA Projectile.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedure and/or equipment designs for approval - See MIL-A-48078. Submit equipment designs, as required, to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine accepeability.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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